

Exercise 7

Implement the simulated annealing algorithm to solve the travelling salesman problem.

Part 1

- Select n points in the plane,
- Use the Euclidean distance between the points as the cost function.
- Generate proposals by swapping two randomly selected points on the route:

For $(1, 2, 3, 4, 5)$ propose e.g. $(1, 2, 5, 4, 3)$ or $(4, 2, 3, 1, 5)$

For $(3, 2, 5, 1, 4)$ propose e.g. $(4, 2, 5, 1, 3)$ or $(3, 5, 2, 1, 4)$

- Apply a cooling scheme, for example:

$$T_k = \frac{1}{\sqrt{1+k}}$$

$$T_k = \frac{1}{\ln(2+k)}$$

- Plot the resulting route in the plane. As a sanity check, test the algorithm using points located on a circle.

Exercise 7 (continued)

Part 2

- Modify the program to work with general travel costs.
- Apply the algorithm to the cost matrix found on Learn.
- Experiment with different design choices, such as cooling schedules and proposal mechanisms.